

Question number	Answer	Mark
6(a)(i)	The only correct answer is A B is not the correct answer as tRNA does not identify a base and transport it to a ribosome C is not the correct answer as tRNA does not form a template for DNA polymerase D is not the correct answer as tRNA does not form a template for RNA polymerase	1

Question number	Answer	Mark
6(a)(ii)	The only correct answer is A B is not the correct answer as ribosomes do not process cytokine proteins C is not the correct answer as the smooth endoplasmic reticulum does not process cytokine proteins D is not the correct answer as vesicles do not process cytokine proteins	1

Question number	Answer	Additional guidance	Mark
6(b)	An answer that includes two of the following points: • adds complementary nucleotides/bases (to template strand) (1) • reference to phosphodiester bond formed (1) • proofreads the DNA (1)	Accept description of formation of H bonds between bases in old and new strand.	2

Question number	Answer	Additional guidance	Mark
6(c)	An explanation that includes three of the following points: • (activation) results in {production of / activation of} transcription factors (1) • (transcription factors) binds to the promotor region (of/for cytokine genes) (1) • (cytokine genes) are transcribed / mRNA produced (1)	ACCEPT secondary messenger systems are activated ACCEPT description of transcription	3

Question number	Answer	Additional guidance	Mark
6(d)	An explanation that includes three of the following points: • cytokines have a specific/ different shape (1) • (cytokines) bind to specific receptors / protein molecule on cell membrane (1) • therefore these cytokines initiate {gene transcription / protein synthesis} in T killer or B cells (1)	ACCEPT tertiary structure ACCEPT cytokines only act on cells with complementary receptors/ complementary binding sites ACCEPT reference to signalling cascade	3

Question number	Answer	Additional guidance	Mark
6(e)	An answer that includes three of the following points: • gene for CD45 protein transcribed into pre mRNA (1) • {spliceosome / enzyme} remove {non coding regions / introns} from (pre) mRNA (1) • translation of (different) mRNA molecules (1) • to produce CD45 protein with a different primary structure (1)	accept ref to post transcriptional modification Do not accept reference to DNA splicing accept rearrangement of exons to produce mRNA accept different 3D/ tertiary structure	3